# Arms Race:

# A Two-Country Model of National Military Spending

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Math 3MB3: Introduction to Modelling

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# 1 Introduction

In Political Theory, specifically in International Relations, the "security dilemma" arises from countries' fear of being attacked, as countries do not know each other's intentions (Tang, 2009). This occurs because there is no higher authority that can govern the interactions between countries and protect them from each other. Suppose there are two countries. One country increases military and armament spending to protect themselves from a potential threat. Although this spending is intended to be defensive, it can make the other country feel threatened because these military and armaments can also be used in an offensive manner. Since the other country does not know how the military and armaments are intended to be used, they also spend on military and armaments to protect themselves. This creates a vicious cycle and can keep escalating until an arms race and possibly a war starts (Tang, 2009). An arms race is when countries compete to increase the quantity or quality of their military and armaments (Gray, 1971). Thus, the security dilemma is called a "dilemma" because countries' actions to protect themselves can ultimately make themselves less secure.

There are several factors which can either mitigate or exacerbate the security dilemma. As the geographic distance between two countries increases, they perceive each other's military spending as less threatening. The type of military technology being invested in also matters. If it is easier for the technology to be used for defensive rather than offensive purposes, then the military spending is perceived as less threatening (Jakobsen & Halvorsen, 2018). On the other hand, misperceptions can cause fear and military spending in both countries to escalate (Tang, 2009).

Currently, the security dilemma is being used to explain the conflict between Russia and Ukraine, between Israel and Palestine, and between Thailand and Cambodia (Marolov & Galeva, 2026). All three conflicts share mistrust and misperceptions as inciting factors. It is important to study what conditions affect the security dilemma to prevent future arms races and avert unnecessary wars.

In this report, we will investigate the following question: ***How does military spending in two countries vary over time based on their interactions?*** We will examine the conditions that lead to an arms race between two countries and the conditions that can maintain peace between them. First, we present linear models that account for mutual fear between the countries, the cost to the economy of increasing military spending, and other political factors that influence how the countries perceive each other's actions. We find that in the long-term there are conditions under which there is no military spending in both countries, conditions where military spending grows without bound, and conditions where military spending reaches a stable level.

In Section 3, we present a logistic model that accounts for the fact that countries have a budget for military expenditure. We find that military spending in both countries stabilizes at a positive value. Then, in Section 4 we model military spending in the United States and Canada by calibrating the parameters in the logistic model to past data. We predict that military spending will slightly decrease in the United States in the future and will slightly increase in Canada. Lastly, we discuss the implications of our models and present directions for future research in Section 5.

## 2.1 Introduction of Base Model and Richardson Model

### Model 1: Base Model

To explore the most simplistic scenario of an arms race, we introduce a two-country military spending model with hypothetical countries: Purple and Green. Furthermore, it is assumed that a country's military spending is governed by the fear of the other country and limited by the country's own economic constraints. With two countries, and two set of parameters, we introduce a continuous model that is a simplified version of the Richardson Arms race model (Olinick,n.d.).

Note, that a continuous model is preferred over a discrete model since before or during conflict, economic constraints, sense of fear and spending on arms and levels of armament fluctuate *instantaneously* rather than at discrete times. Mathematical analysis is also easier with a continuous model especially one that captures an ever evolving and complex real-world scenario.

We assume the following about the two countries

1) Each country's armament rate is directly proportional to the other country's arm spending.
2) There exists a sense of mutual fear, i.e., the more one nation arms, the more the other nation is incentivized to arm.
3) There exists internal economic constraint. Excessive armament expenditure creates an opportunity cost for that country's economy. Thus, each country will reduce its expenditure proportional to its current armament level.

Using differential equations, we obtain the following continuous model:

$$\frac{dp}{dt} = ag(t) - cp(t)$$

$$\frac{dg}{dt} = bp(t) - dg(t)$$

Where:

- $p(t)$ represents the total armament spending of country Purple at time $t$
  - It is measured in country Purple's currency and hence is positive, i.e., $p(t) > 0$

- $g(t)$ represents the total armament spending of country Green at time $t$
  - It is measured in country Green's currency and is positive, i.e., $g(t) > 0$

- $a$ and $b$ represent the mutual fear (expressed in terms of per unit time)
  - $a$ represents Purple's reaction to Green
  - $b$ represents Green's reaction to Purple
  - $a > 0$ and $b > 0$ since we cannot have "negative" fear

- $c$ and $d$ represent the internal economic constraint (expressed in terms of per unit time)
  - $c$ represents country Purple's constraints
  - $d$ represents country Green's constraints
  - $c > 0$ and $d > 0$ since negative values imply that a country always wishes to increase its armament level

## Model 1: Model Solution

Since Model 1 is a homogenous system of differential equations, it can be solved explicitly where:

$$\frac{d\vec{x}}{dt} = A\vec{x} \quad \text{where } \vec{x}(t) = \begin{bmatrix} p_t \\ g_t \end{bmatrix}, \; A = \begin{bmatrix} -c & a \\ b & -d \end{bmatrix}$$

Thus, the general solution to Model 1 is expected to have the form:

$$\vec{x}(t) \; = \; e^{\mathrm{A}t}\,\overrightarrow{x_0} \; = \; Pe^{\mathrm{D}t}\,\overrightarrow{x_0}$$

Where:

- $e^{Dt} = \begin{bmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_2 t} \end{bmatrix}$
- $A = PDP^{-1}$
- $D$ is the diagonal matrix with eigen values $\lambda_1, \lambda_2$
- $P$ is the matrix of the respective eigenvectors
- $\overrightarrow{x_0} = \begin{bmatrix} p_0 \\ g_0 \end{bmatrix}$ where $p_0$ represents the initial armament levels of country Purple and $g_0$ represents initial armament levels of country Green.

In section 2.2, we will use this model solution with educated parameter estimations to analyze stability of equilibria. Further, we will discuss how the eigenvalues of matrix $A$ determine long-term stability.

## Model 2: Richardson's Arms Race Model

We will now consider an extension of Model 1, by accounting for the underlying factors that can influence conflict regardless of time or expenditure level. Upon adding these constants to our existing model, we obtain a model known as the Richardson's arm race model (Olinick, n.d.). These underlying factors will be captured by $e$ for country Purple and $f$ for country Green.

Similar to Model 1, this model remains continuous and relies on differential equations. However, since we add $e, f$ as constants the system is now non-homogeneous, and we will not be able to find an explicit solution; we will rely on numerical methods for further analysis in Section 2.2B.

Hence, Model 2 is now defined as:

$$\frac{dp}{dt} = ag(t) - cp(t) + e$$

$$\frac{dg}{dt} = bp(t) - dg(t) + f$$

Where $a, b, c, d$ and $p, g$ are as defined as in Model 1.

The parameters $e$ and $f$ have the following properties:

- They represent the goodwill or grievances Purple and Green can have towards each other independent of time or expenditure level.
- It also can capture psychological factors like misperceptions, nationalism, perceived threat which can all influence conflict which is independent of time or expenditure level.
- Based on its description it is the only parameter in the model that can take negative and zero values.
    - Positive values represent underlying hostility or ill will one country feels towards the other
    - Negative value represents good will felt towards the other country
    - Hence, the range of the parameters is: $-\infty < e < \infty, -\infty < f < \infty$

Therefore, in contrast to Model 1 which only depends on the magnitude of $a, b, c, d$. Model 2 depends on the magnitude of $a, b, c, d, e, f$ and the sign of $e, f$. We will obtain numerical solution and simulate the various cases in section 2.2B.

## 2.2A: Analysis of Model 1

This section analyzes Model 1. We determine the equilibrium points, analyze their stability, and use simulations to understand the long-term behavior of the model.

### Equilibrium

We explore the long-term behavior of Purple and Green's expenditure levels by finding equilibrium points. This will be the level at which armament expenditure of each country will stabilize with respect to time.

Rewriting our equations in matrix form we obtain:

$$\frac{d\vec{x}}{dt} = A\vec{x}, \qquad \mathrm{A} = \begin{bmatrix} -c & a \\ b & -d \end{bmatrix} \begin{bmatrix} p \\ g \end{bmatrix}$$

At equilibrium, the system satisfies:

$$A\overrightarrow{x*} = 0$$

This yields two equations as follows:

$$-cp^* + ag^* = 0$$
$$bp^* - dg^* = 0$$

Solving the above equations, we obtain the following equilibrium points:

$$(p^*, g^*) = (0,0)$$
$$ab = cd$$

The equilibrium point $(p^*, g^*) = (0,0)$ represents the case of mutual disarmament.

In this case countries Purple and Green are considered allies. There is no incentive to spend based on the others arms level.

This also implies that there is less mutual fear, i.e., the values $a, b$ are small.

This is a very realistic scenario as strong allies often exist where countries are geographically close and have significant economic ties. In such cases, conflict is unlikely due to its negative impact on both.

Alternatively, the equilibrium points $ab = cd$ represent a case of balanced competition:

Responses of Purple and Green are perfectly balanced, the incentive to increase expenditure based on the other's arm level is offset by economic constraints.

There is no single stable outcome, but an infinitely many equilibria that yield a line of equilibria.

Any solution along the line $g = \frac{c}{a}p$ is possible (obtained by solving two equations from above when ab=cd).

Realistic scenario where there is a balance of power and neither side has the incentive to escalate or fully disarm. The system settles on constant level of expenditure, which depends on the initial conditions, with Purple and Green maintaining proportional spending.

## Stability Analysis

While the stability of the equilibrium can be determined through the eigenvalues of the Jacobian matrix $A$, for efficiency, we will use trace and determinant analysis.

Using standard definitions:

$tr(A) = -(c + d)\ ,\ \ det(A) = cd - ab,\ \ \ \Delta(A) = (c - d)^2 + 4ab$

Observe that we defined $a, b, c\ , d\ >\ 0$, hence

- $tr(A) = -(c + d)\ <\ 0$ always holds
- Stability depends on the behaviour of $det(A)$

This yields us the following three cases:

**<u>Case 1: $det(A) > 0\ \ \Rightarrow\ \ cd > ab$</u>**

- $det(A) > 0\ ,\ tr(A)\ < 0\ \ =>$ stable node
- Long-term, economic constraints overpower the sense of mutual fear.
- Respective countries are unable to spend further on military.
- Eventually, as economic constraints continue to overpower this leads to mutual disarmament and peace.
- In context of this model, Purple and Green are more influenced by their internal economic constraints than by their reaction to each other's military spending.
- An example would be, Purple may be facing budget limitations/ fiscal pressures that prevents it from increasing military spending, and similarly Green may prioritize economic stability over military expenditure. So, even though both countries respond to each other, the costs of maintaining these expenditures limits escalation.
- Over time, Green and Purple gradually decrease their military expenditure leading to mutual disarmament and a stable outcome.

Hence the system is stable and approaches $(p^*, g^*) = (0,0)$

Figure 1 simulates this scenario and while parameter choices are arbitrary values, they are chosen such that long-term, economic constraints overpower mutual fear. As a result, in this simulation, we see both countries gradually disarm, and long-term, their military spending settles at 0. Hence, long-term, countries Purple and Green are in an unreactive state with each other.

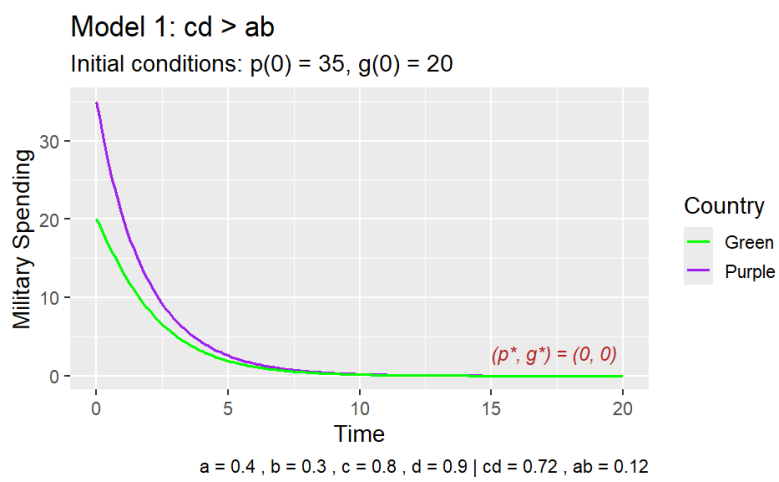


**Figure 1 : Long-term Mutual Disarmament of Countries Purple and Green**

**<u>Case 2: $det(A) < 0 \Rightarrow cd < ab$</u>**

- $det(A) < 0$, $tr(A) < 0$ results in a saddle node; an unstable system.
- Long-term, mutual fear is greater than economic constraints. Each country is highly reactive towards the other.
- Military spending increases over time
- Now Purple and Green's fear from one another outweighs economic limitations. When Green increases spending Purple responds aggressively and vice-versa.
- Creating a positive feedback loop, where there is a continuous escalation of armament levels between Purple and Green.
- While this case suggests unbounded growth in expenditure, in reality, economic constraints as well as pressures from other countries and international organizations like UN would likely limit such escalation.

Here the system is unstable and moves away from equilibrium.

Figure 2 simulates this scenario, parameter choices are arbitrary values, they are chosen such that long-term, mutual fear overpowers economic constraints. From this plot, we see that both countries are stuck in a runaway positive feedback loop; over time, each country increases their military spending due to mutual fear of the other country. As a result, long-term, each country's spending approaches infinity which as mentioned, is unrealistic in practice.

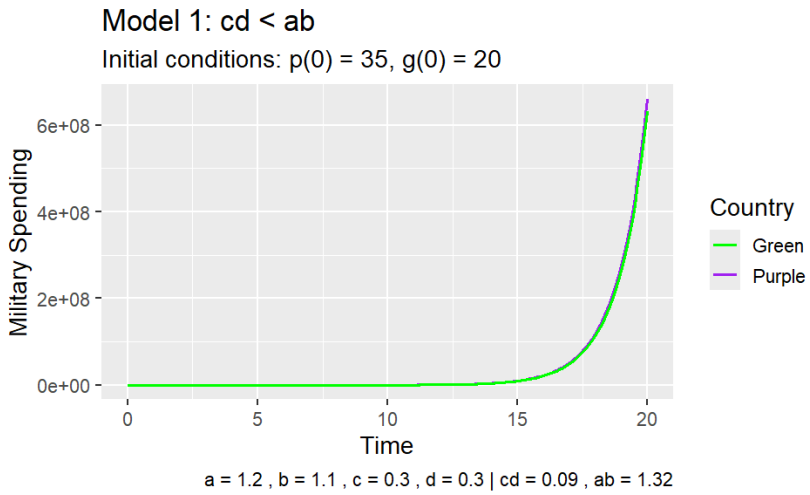


**Figure 2 : Long-Term Escalating Arms Race of Countries Purple and Green**

**<u>Case 3: $det(A) = 0 \Rightarrow cd = ab$</u>**

- This results in a line of equilibria, stable system.
- Long-term, mutual fear and economic constraints perfectly offset each other
- There is no further change in expenditure and countries remain in a neutral state
- Armament spending will settle at some fixed level $(p^*, g^*)$
- Purple and Green are in a perfectly balanced situation where each country's reaction to one another is offset by its economic constraints.
- This leads to stable balance where countries maintain constant level of armament.
- This case most closely reflects real-world behaviour, where countries maintain stable military spending, rather than complete disarmament or unbounded escalation
- For example, many developed nations increase military spending at a stable rate over time, reflecting broader economic growth rather than their relationship with other countries. As seen in the SIPRI data for countries like USA and Canada, military spending as a share of GDP remains approximately stable, indicating a balance between security concerns and economic constraints.

Figure 3 simulates this scenario and while parameter choices are arbitrary values, they are chosen such that long-term, economic constraints offsets mutual fear. In this simulation, Country Green's initial spending is less than Country's Purple. As a result, initially, Country's Green mutual fear outweighs economic constraints and thus, its military spending increases. While the opposite occurs for Country Purple. However, in the long-term, economic constraints offset mutual fear and both countries' spending stabilizes at a fixed ratio (line of infinite equilibria with $g^* = 1.5p^*$). The exact spending is determined by both countries initial spending levels and their respective levels of mutual fear and economic constraint levels.

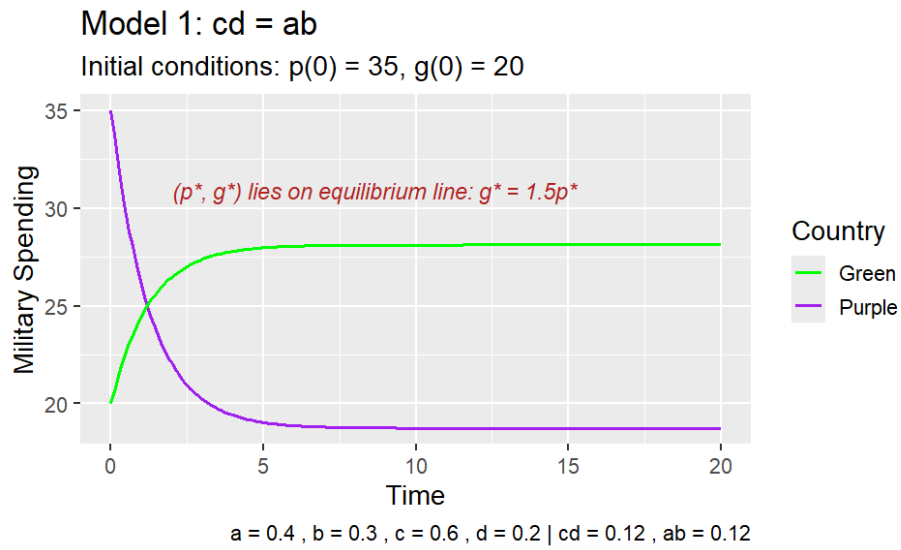


**Figure 3 : Long-Term Neutral Stability of Military Spending of Countries Purple and Green**

## Model Solution

Recall from section 2.1 that the solution of the system can be written as:

$$\vec{x}(t) = e^{\mathrm{A}t}\,\overrightarrow{x_0}$$

This implies that the behaviour of the system depends on the eigenvalues of matrix A and the initial conditions ($\overrightarrow{x_0}$). Noting that:

- If both eigenvalues are negative, military spending decreases over time and approaches zero
- If both eigenvalues are positive, military spending grows over time and approaches infinity (state of arms race)
- If the Det(A) is zero, system converges to line of equilibria depending in the initial conditions

In this model, the explicit solution was used to generate the simulations in R, this helped us show the model behavior under different values of the parameters.

In the R code we specify the parameters as they are fixed, and a specific time interval we want to simulate for. We pass the parameters into a function which arranges them in a 2x2 matrix, then we initialize our output with an empty matrix with 2 rows for the two values and of length of the time interval. Then we iterate starting at some initial values we picked for points, with the next step being described by applying the above equation.

## 2.2B: Analysis of Model 2

This section analyzes Model 2. We determine the equilibrium points, analyze their stability, and use simulations to understand the long-term behavior of the model.

### Equilibrium Points:

Setting the differential equations defined for Model 2 above to zero to find its equilibrium points:

$$ag^* - cp^* + e = 0$$

$$bp^* - dg^* + f = 0$$

This yields us one equilibrium point:

$$(p^*, g^*) = \left(\frac{af + de}{cd - ab}, \frac{cf + be}{cd - ab}\right), \text{ where } ab \neq cd$$

- The non-zero equilibrium shows that Purple and Green settle at positive level of armament spending in the long run.
- This shift is observed after adding the parameters $e$ and $f$, which represents factors like ill-will, goodwill, political tension, and psychological pressures between countries.
- Purple and Green maintain a baseline military spending due to external factors, not just reactions to each other, this is a more realistic reflection of the real world.

### Stability Analysis

For Model 2 the equation can be revised as:

$$\frac{d\vec{x}}{dt} = A\vec{x} + \vec{c} = \begin{bmatrix} \frac{dp}{dt} \\ \frac{dg}{dt} \end{bmatrix} = \begin{bmatrix} -c & a \\ b & -d \end{bmatrix} \begin{bmatrix} p \\ g \end{bmatrix} + \begin{bmatrix} e \\ f \end{bmatrix}$$

Stability continues to depend on $A$ only i.e., it is unaffected by the shift in the equilibrium point caused by e and f.

The same definitions of $tr(A)$, $det(A), \Delta(A)$ continue to hold. The stability depends on the sign of $det(A)$. Hence, we proceed to investigate the same cases and explore what changes under our new model definition.

Any reference to $(p^*, g^*)$ in the below analysis refers to the non-zero equilibrium from above unless stated otherwise. Now we will analyse all the possible cases for Model 2.

**Case 1: $det(A) > 0 \;\Rightarrow\; cd > ab$**

- The system has a stable node, so both Purple and Green's armament levels converge to $(p^*, g^*)$.
- This reflects the situation where economic constraints dominate mutual fear, preventing uncontrolled escalation.
- Across all plots we can see the system moving towards equilibrium, so model prediction matches with our numerical simulations.

All the figures below support our analytical results, showing that Purple and Green converge to equilibrium levels. The parameters were chosen such that economic constraint dominate mutual fear, and then the values of e and f were chosen to explore all the possible case.

**Case 1A: $e, f$ >0**

- Equilibrium: $(p^*, g^*) = (28, 26)$.
- Purple decreases from 35 to 28, Green increases from 20 to 26 and then stabilize. This behaviour is driven by the initial conditions relative to equilibrium: Purple starts above its equilibrium level and decreases, while Green starts below and increases.
- Positive $e$ and $f$ act as constant pressure to maintain armament, raising spending above zero.
- Here, both countries face ongoing political tensions or insecurity, so even with economic constraints, they don't fully disarm and maintain a steady military spending.

In figure 4, we can see Purple and Green converging positive equilibrium levels, where both countries maintain steady military spending due to persistent tension captured by the positive values of $e$ and $f$.

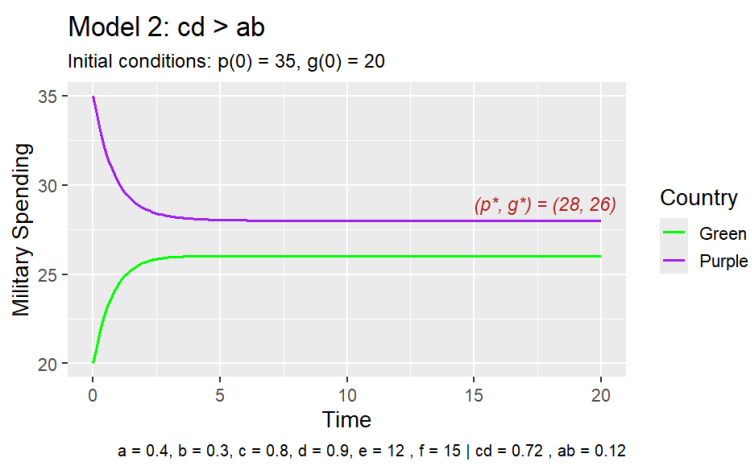


**Figure 4: When cd > ab, stable node exists and $e, f$ variable are strictly positive, leading to long-term spending that is somewhat close**

**Case 1B: $e, f$ <0**

- Equilibrium: $(p^*, g^*) = (-28, -26)$.
- As seen in the plot, both Purple and Green decreases and move towards negative equilibrium values.
- The behaviour still driven by initial condition relative to equilibrium, but equilibrium itself pushed downwards due to negative $e$ and $f$.
- In real-world context, this represents strong incentive for both countries to mutually disarm and cooperate, ultimately settling at $(0,0)$ rather than the negative values predicted by the model, since negative values are unrealistic and highlights a limitation of our model.

In Figure 5, we see Purple and Green converging to their respective negative equilibrium levels; however, negative military spending is not realistic, indicating that this case reflects strong disarmament pressures. The values of e and f were chosen arbitrarily to capture goodwill between Purple and Green.

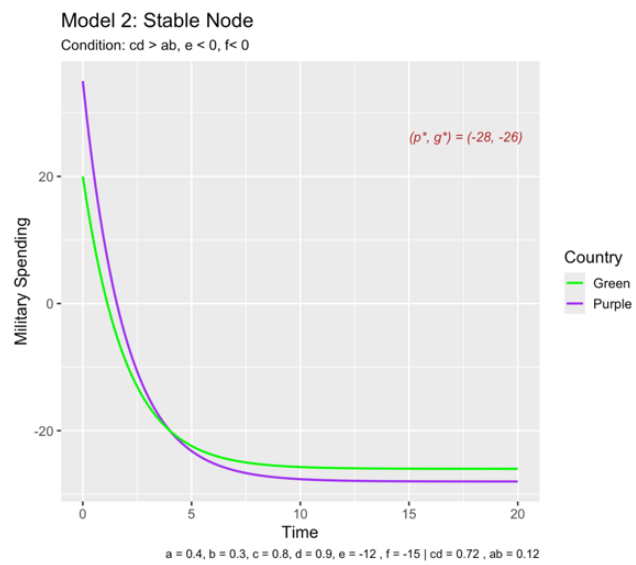


**Figure 5: When $cd > ab$, stable node exists and e,f variable are strictly negative, leading to long-term spending that is somewhat close, as can be seen the behaviour is similar to Case A, however with a shift in equilibrium points.**

**Case 1C: Mixed signs of $e$ and $f$**

- Equilibriums: $(p^*, g^*) = (-8, 14)$ and $(8, -14)$
- One sign being positive and the other being negative causes equilibrium to shift unevenly between Green and Purple.
- In real-world context, the country with negative equilibrium represents a strong push towards disarmament (settling at 0 instead of negative value because that is unrealistic). While the other country continues to maintain military spending due to the ongoing perceived threats.

In figure 6, we see Purple and Green converging to asymmetric equilibrium points, where one country moves towards disarmament while the other maintains positive spending due to the differing

underlying pressure. The values of $e$ and $f$ were chosen to capture imbalance, where one country exhibits goodwill (negative parameter) while the other perceives ongoing threat (positive parameter).

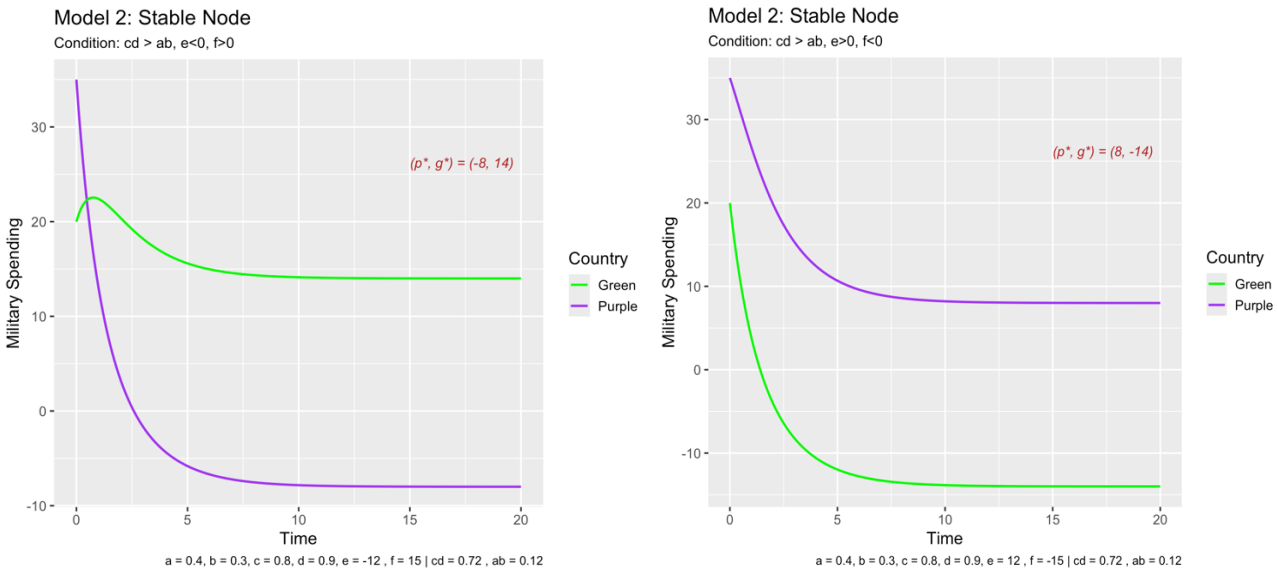


**Figure 6: Flipped signs of e and f, show similar behaviour long-term but with equilibrium values changing sign**

### Case 2: $det(A) < 0 \Rightarrow cd < ab$

- The system has a saddle node, so the system is unstable, therefore not converging at $(p^*, g^*)$.
- This reflects the situation where mutual fear dominates economic constraints, leading to both countries being highly reactive to each other.
- Resulting in the system moving away from the equilibrium, indicating an unbounded escalation or runaway arms race.

All the figures below support our analytical results, showing that Purple and Green move away from the equilibrium over time. The parameters were chosen such that economic constraint is dominated by mutual fear, which leads to highly reactive behaviour between the countries. This results in an unstable system where military spending grows without bound by the end of time interval. The values of e and $f$ were chosen arbitrarily to capture all the possible subcases.

### Case 2A: $e, f > 0$

- No stable equilibrium is observed in the simulation- both Purple and Green's armament level increases over time.
- Strong mutual fear combined with external pressure like political tension, captured by $e = 12$ and $f = 15$, lead to a situation where spending escalates.
- The strong reactions to each other's armament level creates an escalating feedback loop.
- In real world context, this reflects a runaway arms race, where persistent tensions drive both countries to increase spending without stabilization.
- However, such unbounded growth is not realistic due to the economic constraints that exists, highlighting a limitation of this model.

In Figure 7, Purple and Green both increase their military spending rapidly over time, with no stable equilibrium observed. Positive $e$ and $f$ results in continuous escalation because both countries respond strongly to each other's spending. This results in unstable system where spending grows without bound, reflecting a runaway arms race.

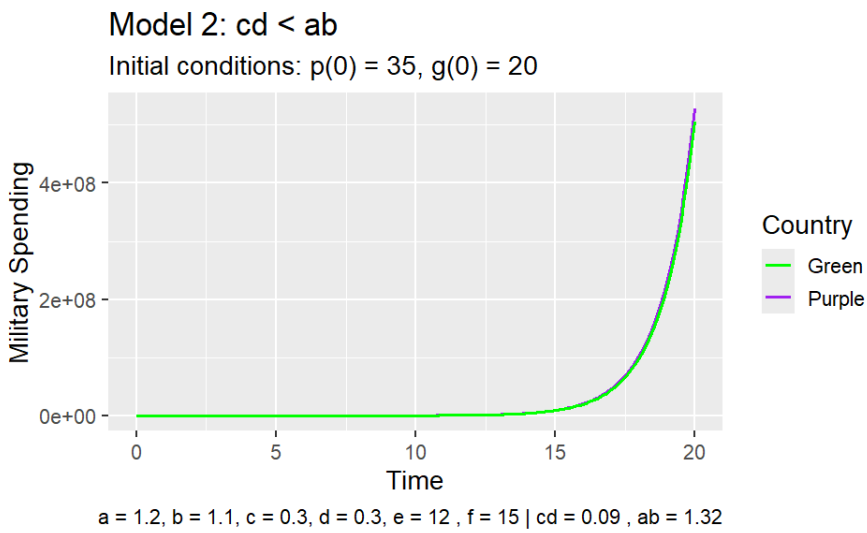


**Figure 7: Unstable equilibrium with positive $e, f$ values exploding at the end of the interval. The $e, f$ values only shift the equilibrium values with respect to Model 1 behaviour**

### Case 2B: $e, f$ <0

- The behaviour is same as case 2A- Purple and Green moving away from equilibrium over time.
- Even though negative $e$ and $f$ represents goodwill between the countries and would indicate that initially the military spending would decrease but the system is unstable.
- Because of this instability, even small increases in one country's military spending could trigger big reactions from the other country leading to escalation over time.
- In real world context, this reflects a situation where goodwill does exist between countries, but mutual fear dominates, which leads to an escalation despite intentions to corporate.

In Figure 8, Purple and Green move away from equilibrium, with military spending increasing rapidly over time. Even though $e$ and $f$ are negative, system remains unstable under condition $cd < ab$. Resulting in spending growing without bound, showing behaviour like figure 7.

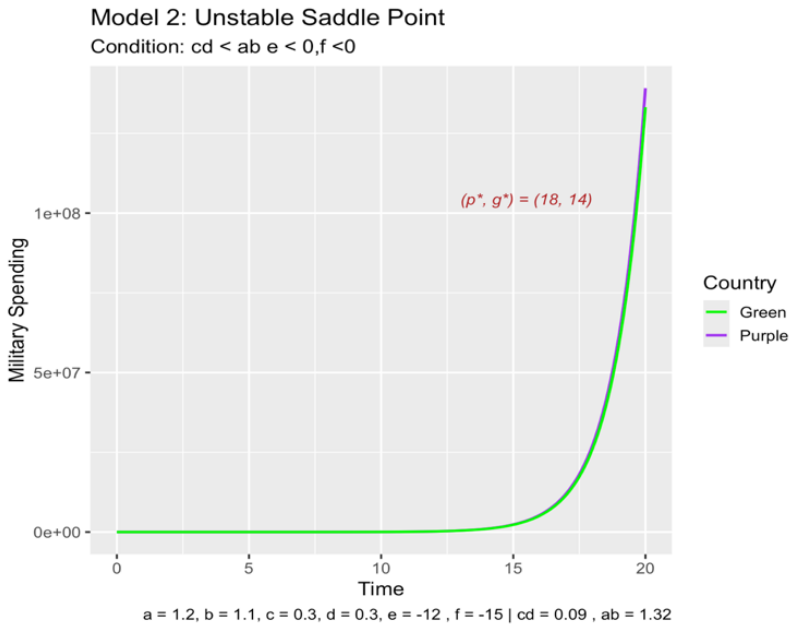


**Figure 8: Unstable equilibrium with negative $e, f$ values exploding at the end of the interval. The $e, f$ values only shift the equilibrium values with respect to Model 1 behaviour as with the prior cases of changing e, f signs in this class of stability**

### Case 2C: Mixed signs of $e$ $and$ $f$

- This case showcases similar behavior to case 2a and 2b- Purple and Green moving away from equilibrium over time.
- Here one country showcases goodwill while the other perceived threat.
- The interaction parameters ($a$ $and$ $b$) dominate the behavior, so for example, even if only Green increases its spending due to perceived threat, Purple still increases its spending in response due to the interaction effect despite there being goodwill from Purple's side
- In real world, this reflects a situation where one country may attempt to cooperate, but strong reactions from the other country lead into arms race.

In Figure 9, Purple and Green move away from equilibrium, with military spending increasing rapidly over time. The behaviour is similar to figure 7 and 8.

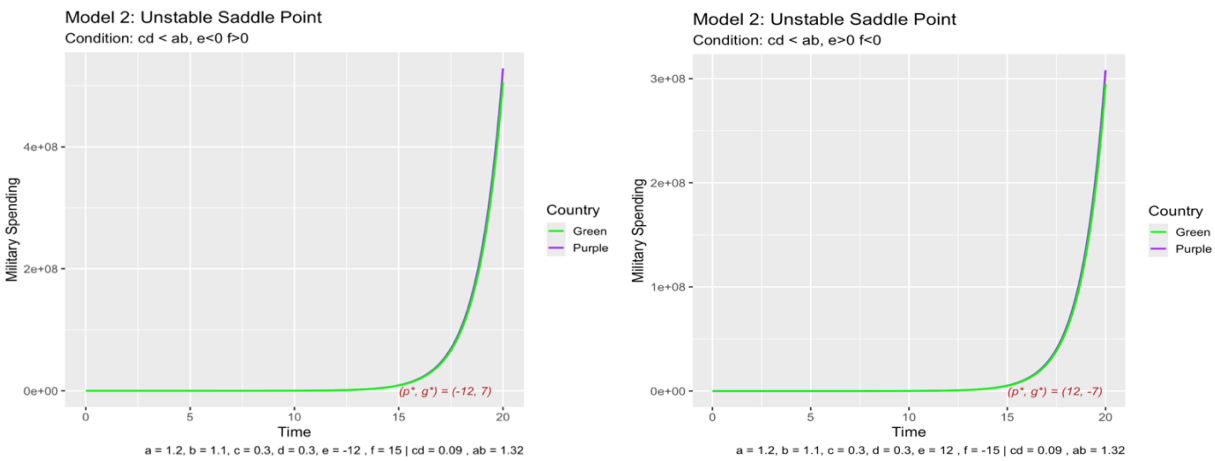


**Figure 9: Unstable equilibrium with differing e, f values exploding at the end of the interval. The e, f values only shift the equilibrium values with respect to Model 1 behaviour as with the prior cases of changing e, f signs in this class of stability**

### Case 3: $\boldsymbol{det(A) = 0 \Rightarrow cd = ab}$

- In this case, mutual fear and economic constraint are exactly balanced.
- This creates a system where behaviour depends strongly on the values of $e$ and $f$.
- If $e$ and $f$ are such that equations are balanced, the system has a line of equilibria in the long run similar to Model 1. The analysis for line of equilibria for case: $cd = ab$:

Equilibrium equations:

$$p = \frac{ag + e}{c}$$

$$p = \frac{dg - f}{b}$$

Rearranging our assumption:

$$\frac{a}{c} = \frac{d}{b}$$

Final equilibrium relationship:

$$\frac{e}{c} = \frac{-f}{b}$$

So, under the case for $cd = ab$, we have an infinite number of equilibrium points given by the above relation for $e/c$ and $-f/b$.

- If e and f are such that the balance is disrupted, the system is no longer stable (no solution in the long run).
- So, stability cannot be determined directly, and we need to dive into the specific subcases to analyse model's behaviour in the long run.
- This represents a fragile situation where small changes in underlying pressures (represented by $e$ and $f$) can significantly alter the long-term behaviour of Purple and Green.

- We can only get a line of equilibria in cases where $e$ and $f$ has mixed signs as it would be impossible to balance the equation if $e$ and $f$ had the same signs.

**Case 3A: $e, f > 0$**

- No equilibrium is observed in the simulation and Purple and Green's military spending increase over time.
- From the plot, the spendings appear almost parallel, indicating the system does not converge to a fixed equilibrium.
- Unlike the other unstable case where growth is rapid, the increase here is more gradual because mutual fear and economic constraints are balanced, so the system does not show strong reactive behaviour.
- However, the constant external pressures (represented by $e$ and $f$) drive a steady and sustained increase in spending, leading to a linear arms race.
- In the real-world context, this reflects a situation where balanced fear and economic constraint prevents rapid escalation, but persistent tensions lead to steady increases in military spending over time.

In Figure 10, Purple and Green increase their military spending steadily over time, with no convergence to an equilibrium. The spending grows almost linearly and indicates no stabilization in the long run. The positive e and f drive continuous growth in the military spending, resulting in linear arms race.

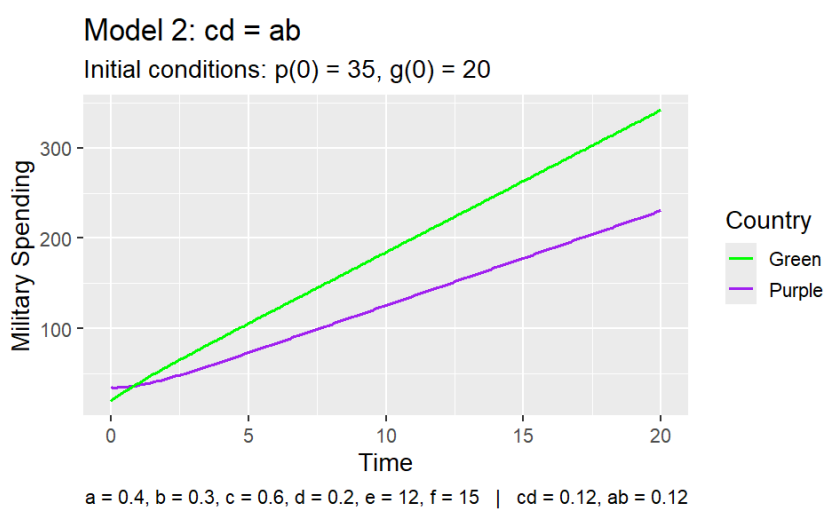


**Figure 10: No equilibrium, escalating arms race**

**Case 3B: $e, f < 0$**

- Behaviour is similar to case 3A - no equilibrium is observed in the simulation
- However, here the spending of Purple and Green decrease overtime because here $e$ and $f$ are negative and therefore represents goodwill.
- The decrease here is gradual because the system does not show strong reactive behaviour making the changes only rely on $e$ and $f$.

- In real-world context, this reflects a situation where goodwill and cooperation lead to countries reducing military spending, but continued reduction is unrealistic because spending cannot go below zero, this is a limitation of the model.

In Figure 11, Purple and Green decrease their military spending steadily over time, with no convergence to an equilibrium. The spending grows almost linearly and indicates no stabilization in the long run. The negative e and f drive continuous reduction in the military spending, resulting in linear de-escalation.

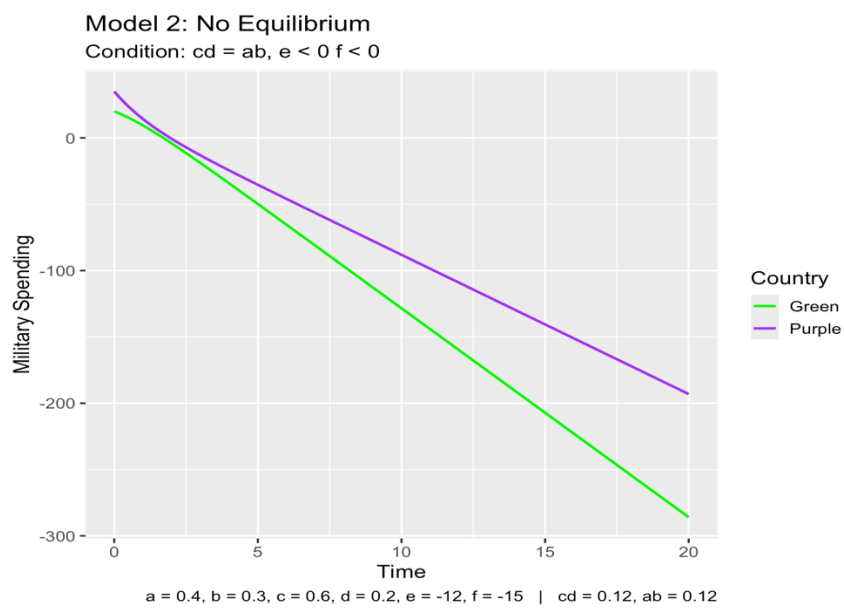


**Figure 11: No equilibrium, negatively escalating arms race**

**Case 3C: Mixed Signs of $e$ and $f$ ($e$ and $f$ don't balance the equations)**

- **$e < 0$ and $f > 0$**
  - $e < 0$ represents goodwill from purple, while $f > 0$ represents that Green perceives threat (ill-will), creating an imbalance of external pressures.
  - Initially, Purple tries to decrease spending due to goodwill while Green tries to increase spending due to perceived threat.
  - Since Purple reacts more strongly ($a = 0.4 > b = 0.3$), its response to Green's behaviour has a larger impact on the system.
  - Therefore, Green's increase in spending triggers a strong reaction from Purple, leading both countries to increase spending, resulting in linear arms race.
  - The system is unstable, so the spending does not converge to an equilibrium and instead diverge over time
  - The long-term behaviour depends on $a$ and $b$ rather than just $e$ and $f$

In Figure 12, Purple and Green diverge over time. With mixed signs of e and f, one country initially reduces spending while the other increases it, but the interaction effects dominate. The system is unstable and increase in Green's spending drives both to escalate over time.

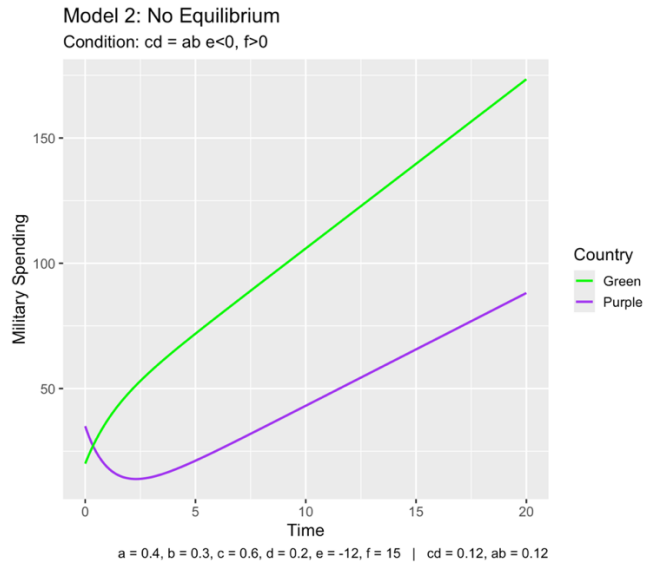


**Figure 12: No equilibrium, escalating arms race when e, f are different signs**

- **$e > 0$ and $f < 0$**
  - Behaviour is similar to the above case- system is unstable and long-term behaviour depends on the country with the stronger reaction parameter.
  - Roles are reversed Purple perceives threat and Green exhibit goodwill.
  - However overall behaviour still depends on interaction term ($a$ and $b$), so Purple's reaction plays a dominant role in the long-term behaviour of the system.
  - Green's reduction in spending triggers a strong reaction from Purple and it decreases its spending, leading to both decreasing their spending over time.

In Figure 13, similar behaviour to Figure 12 but here both countries' spending decreases over time because of reversed sign of $e$ and $f$.

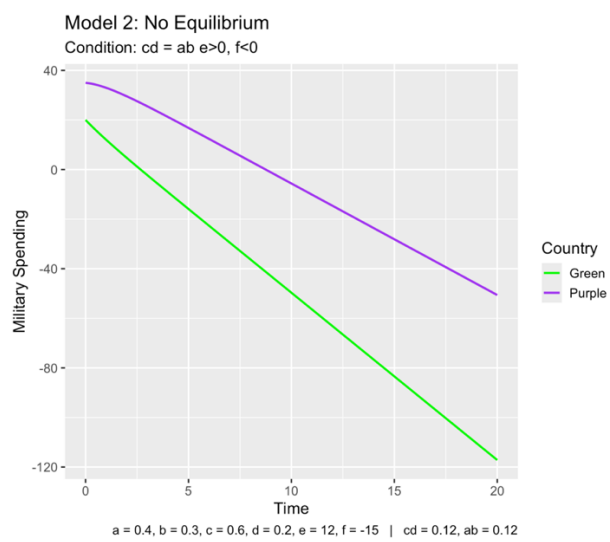


**Figure 13: No equilibrium, negatively escalating arms race when e, f have different signs**

- In real-world context, case 3C reflects the situation where the country that responds more strongly to the other's action drives the overall direction of their relationship in the long run when there is an imbalance in perceived threat or goodwill.

**Case 3D: Mixed Signs of $e$ and $f$ ($e$ and $f$ balance the equations)**

- There is a line of equilibria meaning there are infinitely many solutions.
- The spendings settle along a fixed path, so system settles along this line instead of converging to a point.
- Because the effects of $e$ and $f$ cancel each other out, goodwill or ill-will does not affect system.
- As a result, system behaves similar to the base case of $ab = cd$ ($e = f = 0$) and the outcome depends on initial conditions.
- In real-world context, this case reflects a situation where opposing pressures between countries balance each other out leading to multiple possible stable outcome that depend on the initial conditions.

In Figure 14, the system exhibits a line of equilibria, indicating infinitely many solutions. With $e > 0$ and $f < 0$, the opposing signs balance each other out, and under $ab = cd$, the system becomes stable. The spendings settle along a line of equilibria leading to stabilization depending on initial conditions.

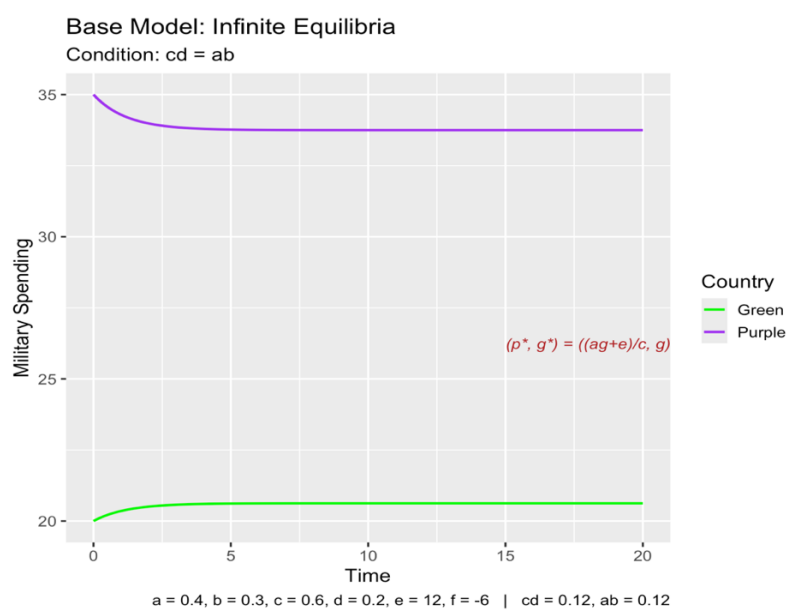


**Figure 14: Infinite Equilibrium points (line), stable equilibrium when e, f and have different signs**

# 3. Extension Model

## Motivation

In the long-term setting where mutual fear overpowers economic constraints, Models 1 and 2 assume an escalating arms race. As mentioned, this scenario involves a positive feedback loop where Models 1 and 2 assume that due to mutual fear, each nation will increase their military spending indefinitely. This highlights the weakness of Models 1 and 2: they allow for unbounded military spending.

To address this limitation, we modified Model 2 by introducing a logistic term involving the military budget limits of each nation, $K_p$ and $K_g$. With the logistic term, mutual fear still increases military spending, but the rate of increase slows as a nation's spending approaches its budget limit. The resulting extension model is shown below:

$$\frac{dp}{dt} = a(1 - \frac{p(t)}{K_p})g(t) - cp(t) + e$$

$$\frac{dg}{dt} = b(1 - \frac{g(t)}{K_g})p(t) - dg(t) + f$$

where:

$a, b, c, d, e, f$ are as defined as in Model 2
$p(t)$: total military spending of country Purple at time $t$, $p(t) \geq 0$
$g(t)$: total military spending of country Green at time $t$, $g(t) \geq 0$
$K_p$: military spending budget limit for country Purple, feasible for $K_p > 0$
$K_g$: military spending budget limit for country Green, feasible for $K_g > 0$

Like Models 1 and 2, this model is multivariate, continuous and dynamical. However, while Models 1 and 2 are linear, the extension model is nonlinear due to the added logistic terms:

$$1 - \frac{p(t)}{K_p}, \qquad 1 - \frac{g(t)}{K_g}$$

Furthermore, while Models 1 and 2 allow for long-term military spending to be unbounded, the long-term military spending is bounded in the extension model. This is because in the extension model, as the military spending of each country approaches their budget limit, the logistic terms slow military spending. Additionally, when military spending exceeds the budget limits, the logistic terms become negative and create a negative feedback loop that decreases military spending.

Hence, when spending of countries Purple and Green, $p(t)$ and $g(t)$, are relatively small compared to their respective budget limits, $K_p$ and $K_g$, the logistic terms are close to 1. In this case, the extension model behaves similarly to Model 2.

However, as $p(t)$ and $g(t)$ approach their respective budget limits $K_p$ and $K_g$, the logistic terms approach 0. In this case, military spending is more influenced by internal cost and grievance or goodwill. Thus, when military spending of each nation approaches its budget limit, $p(t) \rightarrow K_p$, $g(t) \rightarrow K_g$ , the model can be approximated by:

$$\frac{dp}{dt} = -cp(t) + e$$

$$\frac{dg}{dt} = -dg(t) + f$$

On the other hand, if $p(t)$ and $g(t)$ *exceed* their respective budget limits, $K_p$ and $K_g$, it showcases a scenario where countries may prioritize defence over budget deficiency, where being under-defenced is seen as a greater risk than being in budget deficiency. As mentioned, in this case, the logistic terms in the extension model become negative and create a negative feedback loop to decrease spending, preventing unbounded military spending.

## Equilibrium

To find the equilibrium $(p^*, g^*)$ of the logistic model, we set the differential equations equal to zero and solve for $p^*$ and $g^*$. From doing so, the equilibrium can be expressed by the equations below, where the equilibrium $(p^*, g^*)$ can be solved for numerically in `R`[1]:

$$g^* = \frac{K_p(cp^* - e)}{a(K_p - p^*)}, \text{assuming } p^* \neq K_p \text{ to avoid division by 0}$$

$$p^*\left(b - \frac{b}{K_g}\frac{K_p(cp^* - e)}{a(K_p - p^*)}\right) = d\frac{K_p(cp^* - e)}{a(K_p - p^*)} - f$$

## Linearization for Stability Analysis

Since the extension model is non-linear, we apply linearization for stability analysis, where matrix $A$ equals $J(p^*, g^*)$, which is the Jacobian evaluated at the equilibrium $(p^*, g^*)$.

$$\frac{dp}{dt} = a\left(1 - \frac{p(t)}{K_p}\right)g(t) - cp(t) + e = m(p, g)$$

$$\frac{dg}{dt} = b\left(1 - \frac{g(t)}{K_g}\right)p(t) - dg(t) + f = n(p, g)$$

[1] In `R`, the two equations were defined as functions. Since $g^*$ is defined in terms of $p^*$, the `nleqslv` package was used to first numerically solve for $p^*$. Once $p^*$ was solved, it was used to solve for $g^*$.

$$J = \begin{bmatrix} \frac{\partial m}{\partial p} & \frac{\partial m}{\partial g} \\ \frac{\partial n}{\partial p} & \frac{\partial n}{\partial g} \end{bmatrix}$$

$$J = \begin{bmatrix} \frac{-ag}{K_p} - c & a(1 - \frac{p}{K_p}) \\ b(1 - \frac{g}{K_g}) & \frac{-bp}{K_g} - d \end{bmatrix}$$

$$A = J(p^*, g^*) = \begin{bmatrix} \frac{-ag^*}{K_p} - c & a(1 - \frac{p^*}{K_p}) \\ b(1 - \frac{g^*}{K_g}) & \frac{-bp^*}{K_g} - d \end{bmatrix}$$

Since $a, b, c, d > 0$ and $p^* \geq 0$ and $g^* \geq 0$, the signs of the off-diagonal terms of Matrix $A$ depends on whether $p^* \leq K_p$ or $p^* > K_p$, similarly for $g^*$.

Thus, the following about the sign of the terms of matrix $A$ holds:

$$\frac{-ag^*}{K_p} - c < 0$$

$$\frac{-bp^*}{K_g} - d < 0$$

$$a\left(1 - \frac{p^*}{K_p}\right) \geq 0 \; when \; p^* \leq K_p \text{ and } a\left(1 - \frac{p^*}{K_p}\right) < 0 \; when \; p^* > K_p$$

$$b\left(1 - \frac{g^*}{K_g}\right) \geq 0 \; when \; g^* \leq K_g \text{ and } b\left(1 - \frac{g^*}{K_g}\right) < 0 \; when \; g^* > K_g$$

As a result, $tr(A)$ is always negative:

$$tr(A) = \frac{-ag^*}{K_p} - c + \frac{-bp^*}{K_g} - d < 0$$

While $det(A)$ can be either positive or negative:

$$det(A) = (\frac{-ag^*}{K_p} - c)(\frac{-bp^*}{K_g} - d) - a(1 - \frac{p^*}{K_p})\, b(1 - \frac{g^*}{K_g})$$

For this reason, since trace of $A$ is always negative, we conclude that the stability of the equilibrium depends only on the sign of the determinant of $A$. Thus, since $tr(A) < 0$, a stable equilibrium occurs when $det(A) > 0$ and an unstable equilibrium occurs when $det(A) < 0$.

## Results and Discussion: Simulations

Using simulations, we will assess the effect of the logistic term in the extension model by comparing the behaviour between the extension model and Model 2. To do so, we will look at three cases that were analyzed for Model 2:

$$cd > ab, cd < ab, \text{ and } cd = ab$$

<u>When $cd > ab$:</u>

We recall that in Model 2, when economic constraints overpower mutual fear, it reflected the case where the spending of each country stabilizes in the long-term. Figure 15 illustrates that the extension model has the same behaviour.

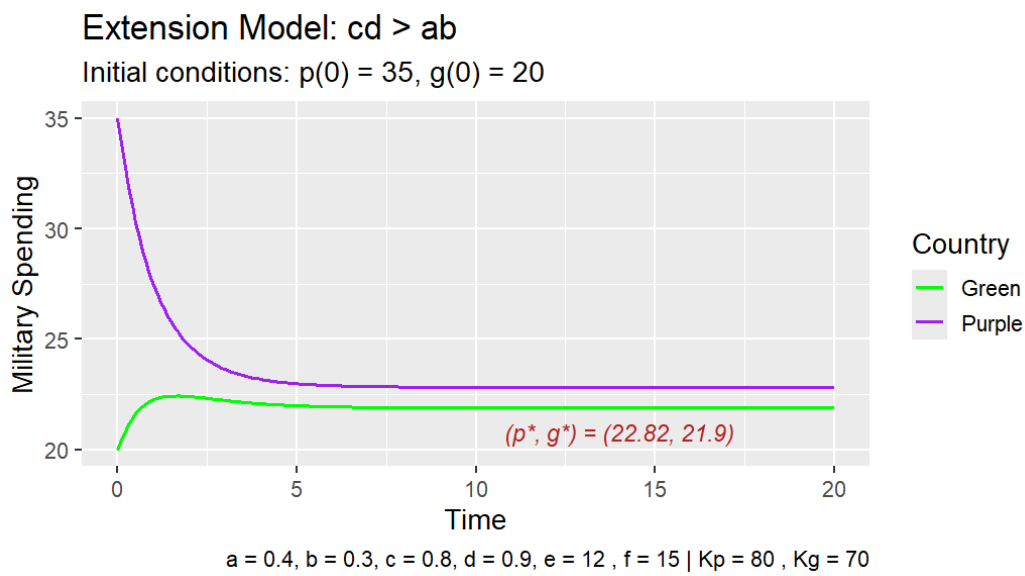


**Figure 15: Extension Model Behaviour when cd > ab**

<u>When $cd = ab$:</u>

We will compare Figures 10 and 16 since they simulate military spending where both models share the same parameter values for $a$ to $f$ and the same initial spending conditions, $p(0)$ and $g(0)$.

In Model 2, the simulation in Figure 10 illustrated that the long-term spending level of each nation did not lead to a unique equilibrium. In contrast, in the extension model, shown in Figure 16, the long-term military spending of each nation stabilizes in the long-term to unique values. While this comparison is on the basis on the specific parameter values and initial spending values chosen, it highlights how Model 2 and the extension model exhibit different military spending behaviours.

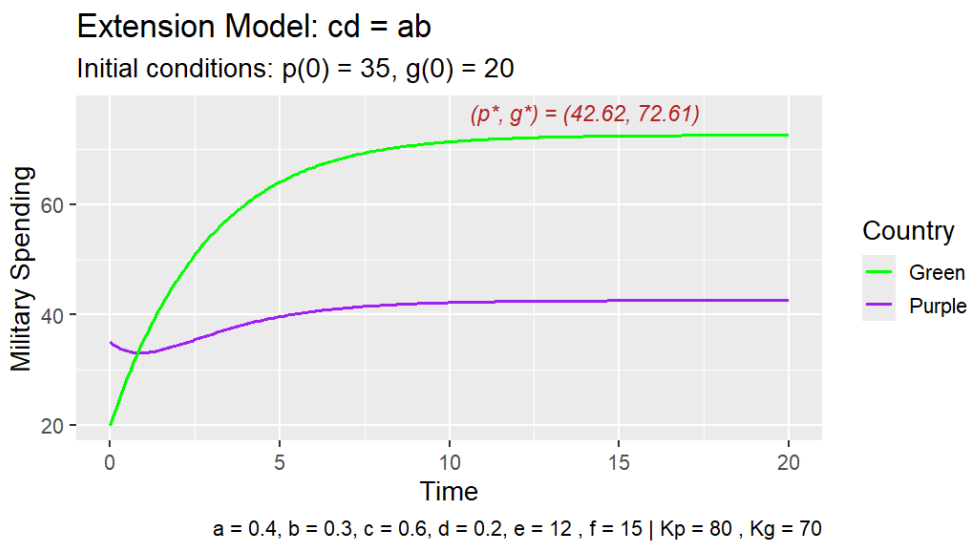


**Figure 16: Extension Model Behaviour when cd = ab**

<u>When $cd < ab$:</u>

We recall that in Model 2, this reflected the case of an escalating arms race. In comparison, as shown in Figure 17, the logistic terms in the extension model successfully limits military spending and addresses the issue of the unbounded military spending we saw in both Models 1 and 2 when mutual fear overpowers economic constraints.

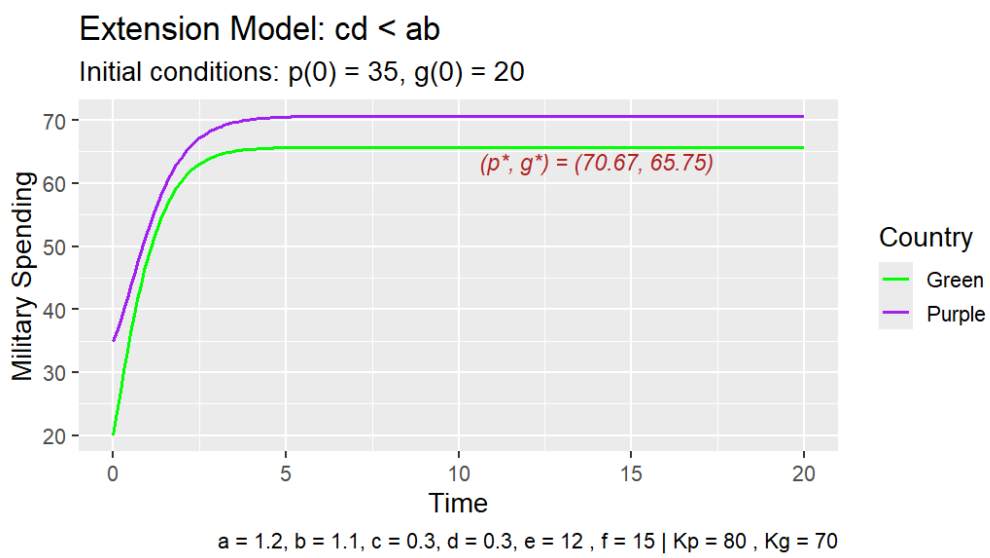


**Figure 17: Extension Model Behaviour when cd < ab**

Now, we will use simulations to understand how the extension model is affected by the initial spending of Countries Purple and Green, $p(0)$ and $g(0)$.

In the below simulations, shown in Figures 18-20, the same parameters are used and so all three simulations share a common equilibrium that was numerically solved to be $(p^*, g^*) \approx (70.67, 65.75)$. This equilibrium evaluated at $det(A)$ is positive, so the equilibrium is classified as stable. Thus long-term, the military spending of each nation stabilizes to a sustainable level.

In Figure 18, we simulate the scenario where initial spending of Countries Purple and Green are less than budget limits, $K_p$ and $K_g$ . In this scenario, the logistic constraint has reduced effect, and the spending of each country is primarily driven by mutual fear. This is seen in the plot, where over time, both countries spending increase due to mutual fear. However, due to the equilibrium being stable, the long-term spending of each nation plateaus and stabilizes to a sustainable level.

From Figure 19, when the initial spending is more than the budget limit, the logistic term constrains spending. Thus, over time, spending *decays* to a sustainable level.

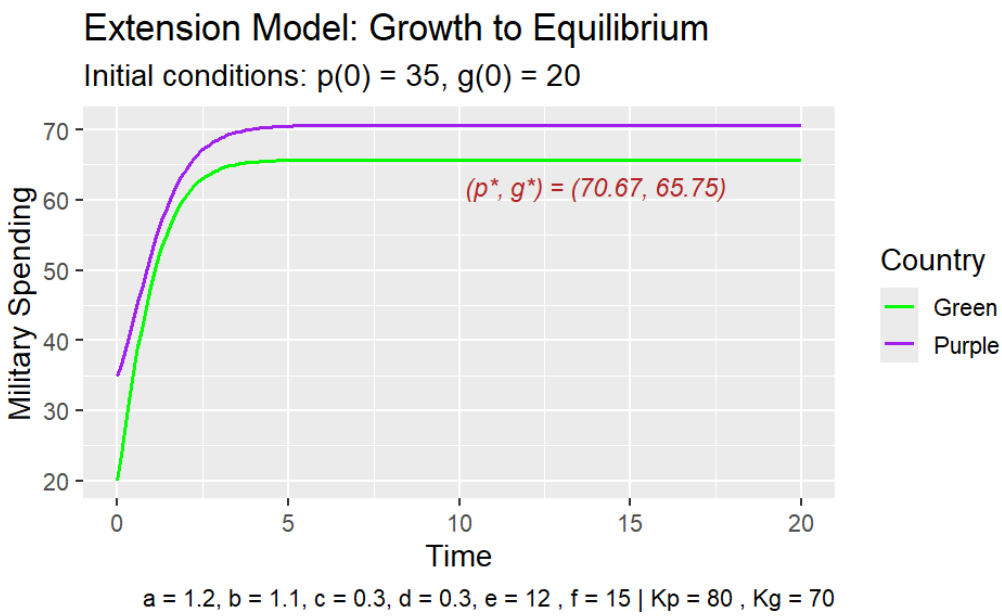


**Figure 18: Extension Model Behaviour when Initial Spending of Countries Purple and Green are Less than their Respective Budget Limits**

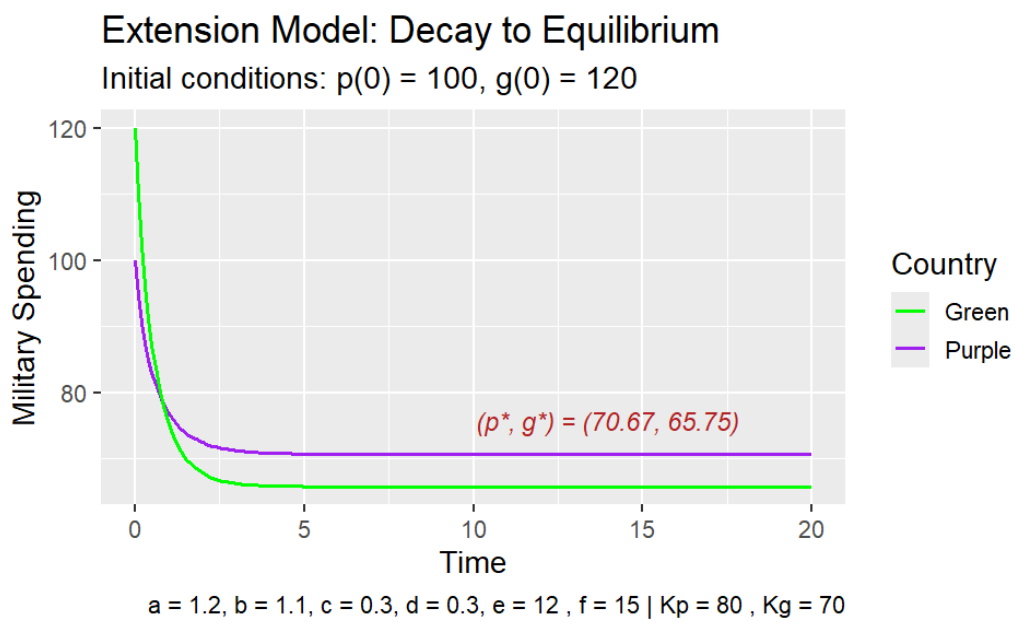


**Figure 19: Extension Model Behaviour when Initial Spending of Countries Purple and Green are Greater than their Respective Budget Limits**

Lastly, from Figure 20, we note that when the initial spending of each country is close to the equilibrium, the initial growth in spending of each nation is slower before stabilizing to a sustainable level.

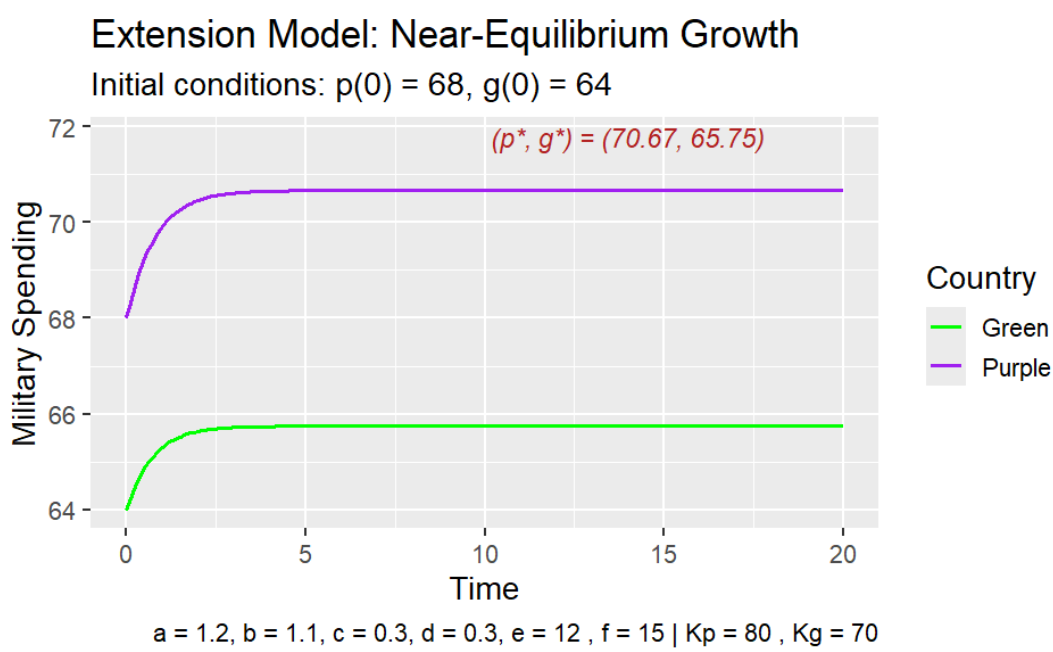


**Figure 20: Extension Model Behaviour when Initial Spending of Countries Purple and Green are close to their Equilibrium**

# 4. Military Spending in United States and Canada

We choose to model how military spending in the United States and Canada influences each other due to their complex relationship. The United States and Canada cooperate in many aspects. They are geographically close, have a strong trade relationship because of the United States-Mexico-Canada Agreement, and are military allies through the North American Treaty Organization (NATO) and North American Aerospace Defense Command (NORAD). However, tariffs imposed by the United States on Canadian aluminum and lumber and competition for resources in the Arctic create tensions between the two countries (Win, 2025).

## 4.1 Calibration Method

To find the parameters $a, b, c, d, e$ and $f$ for the logistic model, we use data on military spending in the United States and Canada from 1950—2024. Military spending is measured in constant 2023 U.S dollars so that spending can be compared between countries without the concern of different currencies being used and spending can be compared over time without the concern of inflation. We use an Euler's method approximation, rather than a continuous model, because our data is measured discretely. We choose the step size h to be equal to one to match the fact that military spending is measured each year, thus t is the current year and t+1 refers to the next year. The equations for the Euler's method approximations are presented below. To keep the notation the same, we assign Canada to be the Purple country and United States is the Green country. We set the initial values $p(0)$ and $g(0)$ to be the level of military spending measured in 1950 for Canada and the United States, respectively. We choose the military spending budget to be about $10,000 USD higher than the maximum value of military spending we observed in the data. For Canada, we set $K_p$ to $40,000 and for the United States, we set $K_g$ to $1,041,000. We find the parameters, highlighted below, that minimize the sum of squared errors between the model and observed data.

Canada

$$p(t+1) = p(t) + h[a\left(1 - \frac{p(t)}{K_p}\right)g(t) - cp(t) + e]$$

United States

$$g(t+1) = g(t) + h[b\left(1 - \frac{g(t)}{K_g}\right)p(t) - dg(t) + f$$

Our logistic model provides a smooth representation of past military spending, which can be seen in Figures 21 and 22. Military spending in both countries follows more of a cyclical pattern rather than a steady upwards increase, therefore a clear limitation of our model is the inability to measure these

peaks and valleys over time. However, we believe our model still provides an accurate long-term estimate.

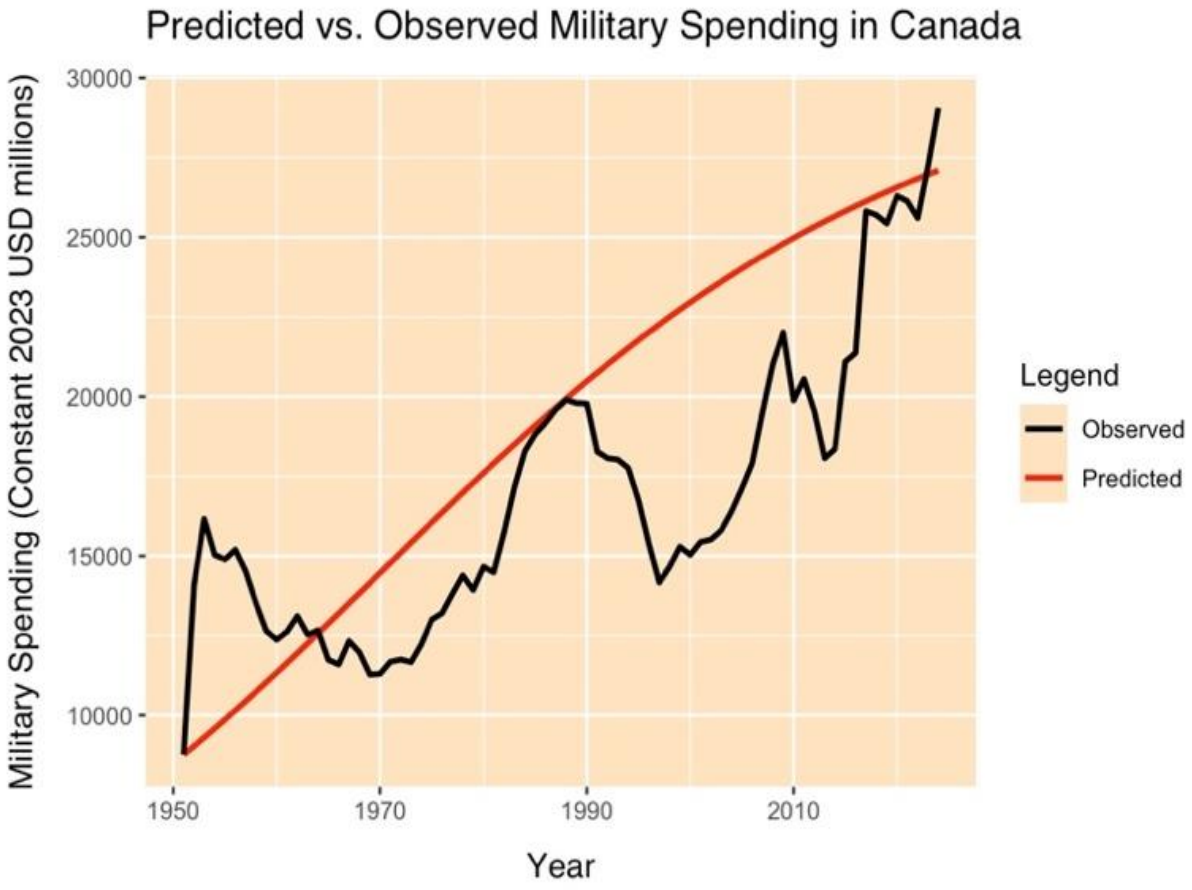


**Figure 21**

For Canadian military spending, we find that $a = 0.001$, $c = 0.006$, and $e = 0.734$. The sign of the parameters $a$ and $c$ match our assumptions. Since $a > 0$, Canada feels threatened by military spending in the United States. Since $c > 0$, the rate of military spending is decreasing as the level of spending increases, reflecting the fact that the government needs to allocate its revenue to provide other services as well. Since $e > 0$, Canada feels enmity towards the United States. This could be because of the tariffs the United States is imposing on Canadian goods and the fact that Canada claims the Northwest Passage in the Arctic as their own waters, while the United States treats it as under international jurisdiction (Win, 2025).

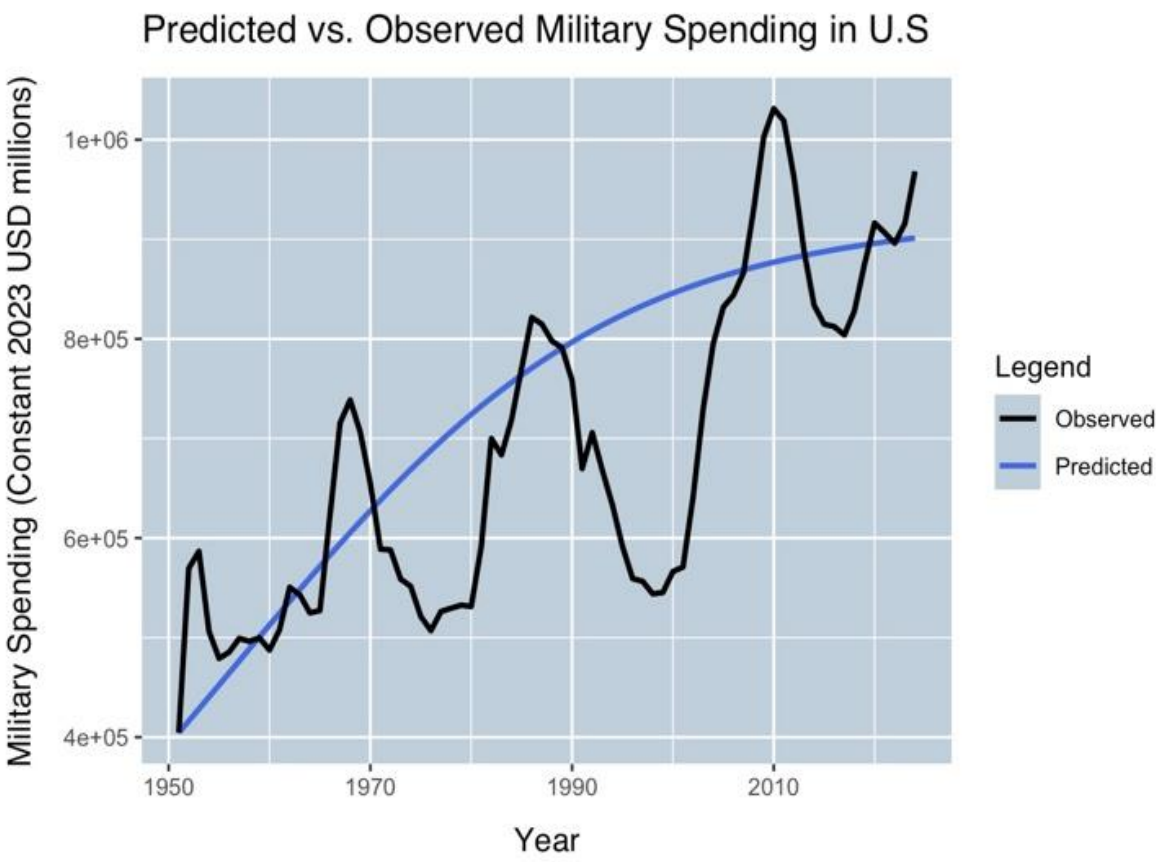


**Figure 22**

For spending in the United States, we find that $b = 3.060$, $d = 0.011$, and $f = -0.598$. The sign of the parameters $b$ and $d$ match our assumptions. Since $b > 0$, the United States feels threatened by military spending in Canada. Since $d > 0$, the rate of military spending is decreasing as the level of spending increases. Although Canada feels negatively towards the United States, $f < 0$, so the United States has a feeling of goodwill towards Canada. This could be because of the shared trade agreements and military alliances between the two countries.

## 4.2 Numerical Simulations

Next, we present numerical simulations to predict what will happen to military spending in the United States and Canada when starting from different initial conditions. We use the calibrated parameters from Section 4.1 and set the step size h to 0.01 to get a more accurate simulation. Thus, t no longer measures years, and the stabilised levels of spending represent long-term behaviour.

First, Figure 23 shows what will happen over time when military spending is initially zero in both countries. We find that military spending stabilises at a value close to each country's military budget in the long-term. Canada reaches 78.4% of its budget ($31,375/$40,000) and the United States reaches 89.3% of their budget ($929,489/$1,041,000).

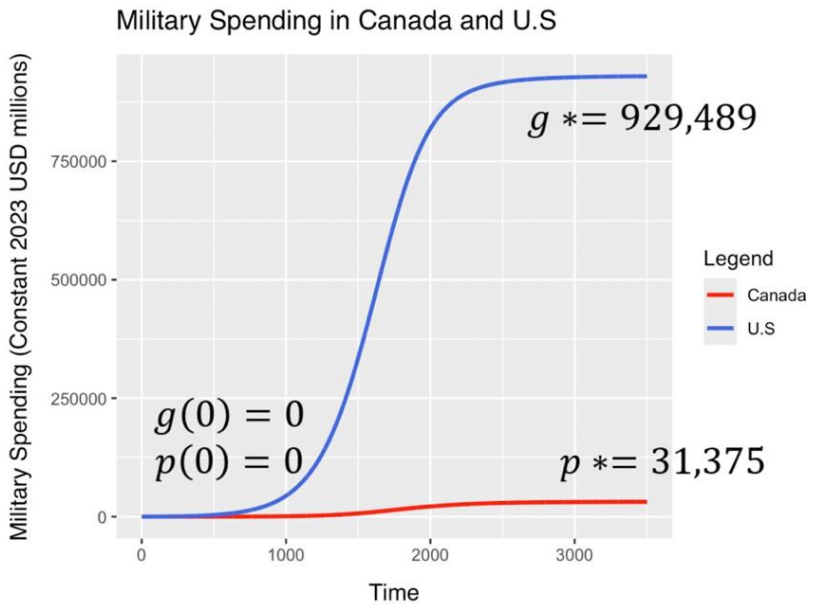


**Figure 23**

We also use the most recent value measured for military spending in both countries, which is from 2024, as the initial condition to predict what will happen to spending in the future. We predict that military spending in the United States will decrease by approximately 4.16% in the future and in Canada it will increase by 5.14%, which can be seen in Figure 24.

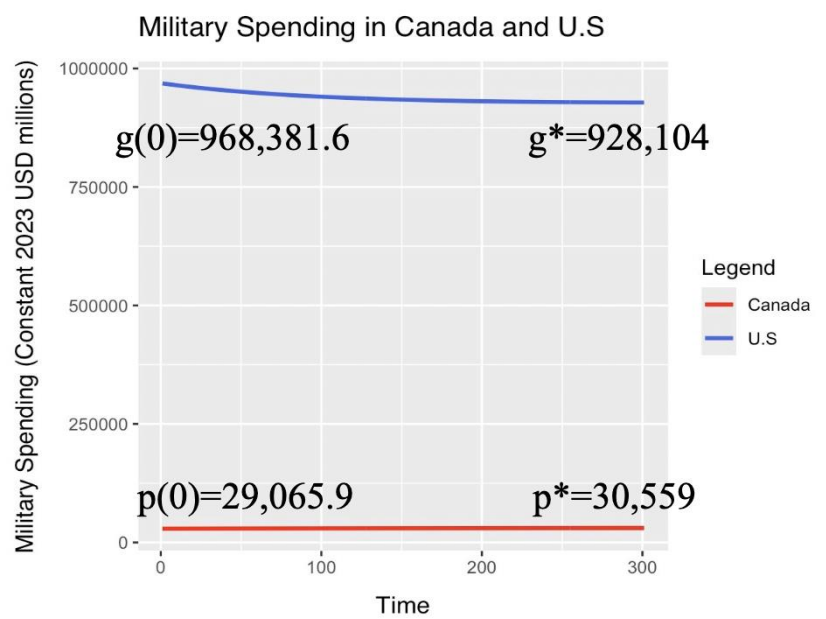


**Figure 24**

## 4.3 Stability Analysis

We classify the stability of the equilibrium points for military spending in the United States and Canada found in the numerical simulations in Section 4.2. As discussed previously, the stability of equilibrium points in the logistic model only depends on the determinant of the Jacobian matrix evaluated at the equilibrium point.

For the first equilibrium point $(p*, g*) = (31\,375, 929\,489)$:

$$det(A) = (\frac{-ag^*}{K_p} - c)(\frac{-bp^*}{K_g} - d) - a(1 - \frac{p^*}{K_p})\, b(1 - \frac{g^*}{K_g})$$

$$= (\frac{-0.001 \times 929489}{40000} - 0.006)(\frac{-3.06 \times 31375}{1041000} - 0.011) - 0.001\left(1 - \frac{31375}{40000}\right) 3.06(1 - \frac{929489}{1041000})$$

$$= (-0.0292) \times (-0.1032) - 0.001 \times (0.2156) \times 3.06 \times (0.107)$$

$$= 0.003013 - 0.00007059$$

$$= 0.0029 > 0$$

Therefore, the equilibrium point that is reached when military spending is initially zero in both countries is stable because det(A) > 0.

For the second equilibrium point $(p*, g*) = (30\,559, 928\,104)$:

$$det(A) = (\frac{-ag^*}{K_p} - c)(\frac{-bp^*}{K_g} - d) - a(1 - \frac{p^*}{K_p})\, b(1 - \frac{g^*}{K_g})$$

$$= (\frac{-0.001 \times 928104}{40000} - 0.006)(\frac{-3.06 \times 30559}{1041000} - 0.011) - 0.001\left(1 - \frac{30559}{40000}\right) 3.06(1 - \frac{928104}{1041000})$$

$$= -0.0292(-0.1) - 0.000236 \times 3.06 \times 0.10845$$

$$= 0.0028 > 0$$

Similarly, the equilibrium point for future military spending in the United States and Canada is stable since det(A) > 0.

# 5. Conclusion

Motivated by the security dilemma, in this report, we investigated how military spending in two countries changed over time based on parameters that characterized their interactions. The Base Model showed that when the economic cost of military spending is a more significant factor than countries' fear of each other, military spending is zero in both countries in the long-term. When mutual fear exactly offsets economic constraints, countries maintain a stable, positive level of military spending in the long-term. When mutual fear is more significant than the costs of military spending, we found that military spending in both countries grows without bound, which is unrealistic.

The Richardson model accounted for other political factors that influence military spending. The model showed that when economic constraints dominate mutual fear, military spending will stabilize at a positive amount if both countries feel negatively towards each other or there will be no military spending if the countries feel positively towards each other. When one country feels good will and the other feels enmity, there will be no military spending in the former and a stable level of military spending in the latter. When mutual fear is more significant than economic constraints, there will be unbounded military spending in both countries regardless of how they feel towards each other. When economic constraints exactly balance mutual fear, if the countries feel negatively towards each other, there will be a steady but not exponential increase in spending over time, and there will be a steady decrease in spending over time when the countries feel positively towards each other.

To address the limitations of our linear models that allowed for an infinite level of military spending in both countries, we introduced the logistic model with a military spending budget for both countries. In the logistic model, we found that military spending stabilizes at some positive level in both countries in the long-term. Using data on the United States and Canada, we found that the logistic model predicts that military spending reaches close to the budget limit in both countries when both countries are initially spending zero. Ultimately, the model does not predict any dramatic changes in military spending in the United States or Canada for the future.

There are limitations to our models. The logistic model assumes that the military budgets $K_p$ and $K_g$ are fixed. In practice, over time, the military budget limits fluctuate due to the political climate and changes in government revenue. Thus, as a future direction, the logistic model could include time-varying military budget limits, $K_p(t)$ and $K_g(t)$, as seen in general logistic models (Meyer & Ausubel, 1999).

Additionally, our models assume that one country knows exactly how much the other country is spending. In reality, there is asymmetric information between countries because countries can make security plans in secret. Thus, when modelling military spending in the Purple country, it would be more accurate to use Purple's prediction of Green's spending in the model rather than Green's actual level of spending and vice versa.

When modelling military spending in the United States and Canada, we noticed that the fear term for the United States was significantly larger than the fear term for Canada. We do not believe that the

United States feels more threatened by Canada. This can be explained by the fact that the United States exhibited much larger changes in military spending over the time than Canada, which is likely because their military spending depended on interactions with other countries, and not just Canada. Therefore, to improve accuracy, spending in several other countries should also be added to the models.

# 6. Appendices

## Individual Contributions

| *Group Member* | *Individual Contributions* |
|---|---|
| *Aadya Mishra* | ▪ Section 2.2 A and B, did the analysis of all cases and subcases using the simulations generated by the other group members<br>▪ Connected the predictions of the cases to real world context<br>▪ Co-worked on setting up base models and the associated mathematical calculations (section 2.1 and 2.2)<br>▪ Presented the cases and simulations of both models in the presentation |
| *Harjot Sanghera* | ▪ Co-worked on research, mathematical calculations, equilibrium/ stability conditions and initial drafting for base and Richardson models presented in section 2.1, 2.2 and appendices<br>- Prepared corresponding presentation slides for 2.1,2.2<br>- Presented section 2.1<br>▪ Edited and finalized section 2.1<br>▪ Contributed towards the conclusion of the report relating to section 2 |
| *Maryam Nainar* | ▪ All contents in Introduction<br>▪ All contents in Section 4<br>▪ Conclusion |
| *Nivetha Nimalanesan* | ▪ Simulations for Model 1<br>▪ Simulations for Model 2 (when both e and f are positive)<br>▪ Simulations for extension model<br>▪ All content in Section 3 on the extension model<br>▪ Potential developments for the extension model found in Conclusion |
| *Pesara Amarasekera* | ▪ Simulation Work mainly for Model 2 (R-code, plots, discussion)<br>▪ Prepared Phase portraits and interpreted results for presentation but cut them out, since we have enough<br>▪ Derivations for lines of equilibria (cd = ab), and help with other derivations as needed<br>▪ Interpretation and discussion of other sections as needed (discussion of extensions, limitations etc.) |

## Calculations

### Model 1

**Finding Equilibrium Points:**

$$\frac{dp}{dt} = ag(t) - cp(t)$$

$$\frac{dg}{dt} = bp(t) - dg(t)$$

Rewriting our equations in matrix form we obtain:

$$\frac{d\vec{x}}{dt} = A\vec{x} = \begin{bmatrix} \frac{dp}{dt} \\ \frac{dg}{dt} \end{bmatrix} = \begin{bmatrix} -c & a \\ b & -d \end{bmatrix} \begin{bmatrix} p \\ g \end{bmatrix}$$

At the equilibrium, armament expenditure invariant with respect to time we represent this using p ∗ and g ∗

$$A\vec{x} = \begin{bmatrix} -c & a \\ b & -d \end{bmatrix} \begin{bmatrix} p^* \\ g^* \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This yields two equations as follows:

$$-cp^* + ag^* = 0 \quad ------ \quad (eqn\ 1)$$

$$bp^* - dg^* = 0 \quad ------ \quad (eqn\ 2)$$

From (eqn 1) we obtain:

$$ag^* = cp^*$$

$$\Rightarrow g^* = \frac{cp^*}{a} \quad \text{------ } (eqn\ 3)$$

Plugging in (eqn 3) into (eqn 2):

$$bp^* - d\left(\frac{cp^*}{a}\right) = 0$$

$$\Rightarrow p^*\left(b - \frac{dc}{a}\right) = 0 \text{ ------ } (eqn\ 4)$$

(eqn 4) yields:

$$p^* = 0 \text{ and } \left(b - \frac{dc}{a}\right) = 0$$

Plugging in $p^*$ into (eqn 3) yields:

$$p^* = 0 \Rightarrow g^* = \frac{c(0)}{a} = 0$$

$$\Rightarrow (p^*, g^*) = (0,0)$$

Similarly,

$$\left(b - \frac{dc}{a}\right) = 0 \Rightarrow ab = cd$$

**Stability Analysis:**

We will explore the stability of each equilibrium point through finding the eigenvalues and classifying its stability.

$$\det(A - \lambda I) = 0$$

$$\Rightarrow \det\left(\begin{bmatrix} -c-\lambda & a \\ b & -d-\lambda \end{bmatrix}\right)$$

$$\Rightarrow \lambda^2 + \lambda(c+d) + (cd - ab) = 0$$

Using the quadratic formula:

$$\lambda = \frac{-(c+d) \pm \sqrt{(c+d)^2 - 4(cd-ab)}}{2}$$

Also recall the following standard definitions:

$$tr(A) = -(c+d)\,,\ \ det(A) = cd - ab,\ \ \Delta(A) = (c-d)^2 + 4ab$$

Hence:

$$\lambda_1 = \frac{-tr(A)+\sqrt{\Delta(A)}}{2} \text{ and } \lambda_2 = \frac{-tr(A)-\sqrt{\Delta(A)}}{2}$$

Observe that we defined $a, b, c, d > 0$, hence

- $tr(A) = -(c+d) < 0$ always
- Stability depends on the behaviour of $det(A)$

**Model Solution:**

We obtain a model solution that will help us understand stability better based on initial points.

$$\frac{d\vec{x}}{dt} = A\vec{x} \text{ where } \vec{x}(t) = \begin{bmatrix} p(t) \\ g(t) \end{bmatrix}, A = \begin{bmatrix} -c & a \\ b & -d \end{bmatrix}$$

General solution to Model 1 is expected to have the form, where $\overrightarrow{x_0} = [\, p_0,\ g_0]$ and $\overrightarrow{x_t} = [\, p_t,\ g_t]$:

$$\vec{x}(t) = e^{At}\overrightarrow{x_0} = Pe^{Dt}\overrightarrow{x_0}$$

Where:

- $e^{Dt} = \begin{bmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_2 t} \end{bmatrix}$
- $A = PDP^{-1}$
- $D$ is the diagonal matrix with eigen values $\lambda_1, \lambda_2$
- $P$ is the matrix of the respective eigen vectors
- $\overrightarrow{x_0}$ represents the initial armament expenditure

This indicates that the long-term behaviour depends on the eigenvalues of $A$ i.e., $\lambda_1, \lambda_2$

- If $\lambda_1, \lambda_2 < 0$ then $e^{\lambda_1 t}, e^{\lambda_2 t} \rightarrow 0$ indicating armament expenditure approaching zero
- If $\lambda_1$ or $\lambda_2 > 0$ then $e^{\lambda_1 t}$ *or* $e^{\lambda_2 t} \rightarrow \infty$ indicating increasing armament and a state of arms race.

Model 2

**Finding Equilibrium Points:**

$$\frac{dp}{dt} = ag(t) - cp(t) + e$$

$$\frac{dg}{dt} = bp(t) - dg(t) + f$$

In a similar procedure to above setting the differential equations equivalent to zero and recognizing that the equilibrium point does not with respect to time we obtain the following equations.

$$ag^* - cp^* + e = 0 \text{ ------ } (eqn\ 5)$$

$$bp^* - dg^* + f = 0 \text{ ------ } (\text{eqn } 6)$$

Re-arranging (eqn 5) yields:

$$g^* = \frac{cp^* - e}{a} \text{ ----- (eqn 7)}$$

Plug in (eqn 7) into (eqn 6):

$$bp^* - d\left(\frac{cp^* - e}{a}\right) + f = 0$$

$$\Rightarrow \text{ab}p^* - dcp^* + de = -af$$

$$\Rightarrow p^* = \frac{-af - de}{ab - dc} \text{ ----- (eqn 8)}$$

Plug in (eqn 8) into (eqn 7) to obtain our first equilibrium point:

$$g^* = \frac{c(\frac{-af-de}{ab-dc}) - e}{a}$$

$$\Rightarrow \frac{-caf - cde - eac + ecd}{a(ab-cd)} = \frac{a(-cf-ec)}{a(ab-cd)}$$

Hence our first equilibrium point is:

$$(p^*, g^*) = \left(\frac{af+de}{cd-ab}, \frac{cf+be}{cd-ab}\right) \text{ ----- (eqn 9) where } ab \neq cd$$

**Stability Analysis:**

The model equation can be revised as:

$$\frac{d\vec{x}}{dt} = A\vec{x} + \vec{c} = \begin{bmatrix} \frac{dp}{dt} \\ \frac{dg}{dt} \end{bmatrix} = \begin{bmatrix} -c & a \\ b & -d \end{bmatrix} \begin{bmatrix} p \\ g \end{bmatrix} + \begin{bmatrix} e \\ f \end{bmatrix}$$

Stability continues to depend on $A$ only i.e. it is unaffected by the shift in the equilibrium point.

The same definitions of $tr(A)$, $det(A), \Delta(A)$ continue to hold. As in Model 1, since tr(A) is always negative under parameter constraints, the stability depends on the sign of $det(A)$.

**Analysis for line of equilibria for case: cd = ab**

$$Assumption : cd = ab$$

$$ag - cp + e = 0 - - - 1a$$
$$bp - dg + f = 0 - - - 1b$$

Rearrange 1a and 1b we get:

$$cp = ag + e$$
$$bp = dg - f$$

Rearranging and solving for p in both cases

$$p = \frac{ag + e}{c}$$
$$p = \frac{dg - f}{b}$$

Rearranging our assumption

$$\frac{a}{c} = \frac{d}{b}$$

Both lines have a slope when by assumption

$$\frac{ag}{c} = \frac{dg}{b}$$

And

$$\frac{e}{c} = \frac{-f}{b}$$

So under the case for cd = ab, we have an infinite number of equilibrium points given by the above relation for e/c and -f/b

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